MOLDOVA – JOINT BANK-FUND DEBT SUSTAINABILITY ANALYSIS[1]

Based on the external LIC DSA, Moldova's risk of debt distress is low. The public DSA suggests that Moldova's overall public sector debt dynamics are sustainable in light of the current size and the evolution of the domestic debt stock.[2]

A. Background

1. **Moldova's debt situation has improved since the last DSA in early 2006**. At that time, short-term financing needs stemmed from the projected financing gaps from external shocks and the stock of arrears owed to bilateral creditors (mainly Russia, the US and Japan). The latter concerns were subsequently resolved through Paris Club rescheduling in May 2006. Debts covered under the agreement were 33 percent of Moldova's total debt stock, based on end-April 2006 numbers.[3] However, the reduction in external debt was not as dramatic as it had been from 2000–2005: external debt was 124 percent of GDP in 2000 (with public debts totaling 95 percent of GDP), falling to 56 percent of GDP in 2005 (27 percent of GDP for public debt).[4]

2. **Prospects for official donor assistance have also improved.** In response to the twin external shocks—doubling of energy prices and Russia's ban on imports of Moldovan wine—large commitments were made by donors at the December 2006 Consultative Group meeting (primarily grants and technical assistance) and are expected to materialize over the next four years. Moldova has also qualified for grants from the Millennium Challenge Corporation, reducing the pressure to borrow for growth-enhancing investments. Finally, the PRGF arrangement was augmented at the time of the first review in December, raising the total amount of access to Fund resources from SDR 80.08 million to SDR 110.88 million.

3. **A significant fraction of external debt is on concessional terms, and will remain so.** Table 1 illustrates the preponderance of debt owed to multilateral lenders on concessional terms and the likely evolution from future disbursements. Most borrowing in the near term is forecast to be on concessional terms, either directly or through integrated loan arrangements that incorporate grants. Over the longer term as Moldova graduates from IDA-only eligibility, borrowing will continue to be primarily from multilateral lenders, but will become less concessional. Loans from commercial creditors make up a very small fraction of the total.

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[1] Data and analysis exclude Transnistria.

[2] The DSA has been produced jointly by the Bank and the Fund, in consultation with the EBRD.

[3] Moldova owed Germany, Italy, Japan, Russia, and the US a total of $274 million, out of $822 million.

[4] The rapid fall in debt to GDP was mainly due to strong GDP growth; the nominal amount of debt remained relatively constant.

Table 1: External Public Debt Profile

	2007		2008-27 2/	
	Total	Concessional 1/	Average	Concessional 1/
Debt stock (US$):	945.8	402.6	944.9	322.7
Multilateral	618.8	402.6	673.5	273.4
World Bank	408.4	272.0	510.2	243.1
IMF	169.0	114.2	64.8	25.3
Others	41.4	16.4	98.5	5.0
Bilateral	304.2	0.0	263.2	49.3
Commercial	22.8	0.0	8.2	0.0
New disbursements (US$):	96.7	88.1	53.7	19.2
Multilateral	93.6	88.1	49	14.5
World Bank	34.2	34.2	36.5	9.0
IMF	49.8	49.8	2.5	2.5
Others	9.6	4.1	10.0	3.0
Bilateral	1.8	0.0	4.7	4.7
Commercial	1.3	0.0	0.0	0.0

1/ Concessional loans have a minimum grant element of 35 percent.
2/ Simple average.

4. **Private debt is growing, with loans associated with FDI becoming more important.** Private external debt can be divided between regular loans (classified as liabilities under other capital in the capital account), loans associated with FDI transactions, and energy arrears. In 2001 new disbursements from FDI-linked loans comprised 35 percent of new borrowing, but by 2006 the proportion was 52 percent. Private debt is also increasingly denominated in euros – 20 percent in 2006 compared with only 6 percent in 2002. This is a welcome trend, given the natural hedge from remittances denominated in euros.

5. **The increase in energy prices led to private external arrears.** There is a large outstanding stock of historic debt from unpaid energy bills (totaling $286 million at end-2005), but in recent years current payments at least have been made. However, while electricity and natural gas tariffs are set at cost recovery, tariffs for heat are below that level causing Moldovagaz to run arrears of about $34 million to Gazprom in 2006, and early indications are that arrears will continue in 2007. Heating tariffs are now set by an independent regulator, and are scheduled to rise to full cost-recovery levels over the course of the program.

B. Underlying Assumptions

6. **Box 1 summarizes the medium-term macroeconomic framework underlying the DSA.** Of particular importance is the assumption that remittances will continue to finance the balance of payments. Moldova has one of the highest proportions of remittances to GDP in the world, 33 percent in 2006. Nominal growth of remittances (in US$) has averaged

40 percent since 2001, reaching a peak of 54 percent in 2003 and subsiding to 29 percent in 2006. Strong growth in the euro area provides incentives to work abroad, but border crossings are now complicated by the accession of Romania to the EU at the beginning of the year (because of new visa requirements). Staff project remittance growth of 20 percent in 2007, and growth sufficient to keep remittances around 30 percent of GDP for the foreseeable future.

7. **The overall fiscal deficit is expected to remain at about 0.5 percent of GDP over the medium term, permitting modest primary surpluses to accumulate**. High levels of remittances should continue to feed import growth, VAT and excises on which will bolster revenue. Recent policies, such as an increase in civil service wages are expected to be offset by a medium-term rationalization of public sector employment. Despite the increase in donor assistance for capital expenditures over the next few years, there is not a large growth dividend built into the baseline scenario. Indeed, growth of 5 percent over the medium term assumes a mild deceleration from the pace of the past few years.

Box 1. Moldova – Macroeconomic Assumptions

The macroeconomic assumptions have not changed significantly since the last DSA: growth will remain strong, inflation to subside, with continuing current account deficits and primary government surpluses.

Real GDP growth is projected to average 5 percent through 2012 and falling to 4 percent thereafter. While this is higher than the historical average, the past decade included two years of severe recession in the aftermath of the Russian crisis of 1998 causing average growth rates to fall and volatility to rise. In the post-crisis period (2001–2006) real growth averaged 6.6 percent.

Inflation is projected to subside to the single-digit range by end-2007, falling to three percent by 2027 (as measured by the GDP deflator). This is considerably lower than the historical average of 16 percent, but measures taken under the program should ensure the National Bank of Moldova keeps a tight rein on monetary policy.

The **current account** is projected to continue to be supported by strong remittances, which in 2006 reached 33 percent of GDP. As remittances largely feed into consumption and imports, changes in flows largely net out with respect to the current account deficit. The improved prospects for the business environment have also raised capital inflows, in particular direct investment. Exports, which were down sharply after the Russian ban on wine, are projected to recover only gradually, as a mechanism for resuming exports has not yet been arranged, despite the formal lifting of the ban.

The **overall fiscal balance** is expected to remain negative, but with a small deficit. Grant assistance from donors, especially the European Union, is projected to increase, as are project loans that flow through the budget. This fiscal stance reflects the need to increase capital investment and social spending over the medium-term.

8. **External public debt and debt-service ratios remain well below the thresholds in the baseline and historical scenarios.** External debt and debt-service to exports ratios are

healthy, despite the continuing effects of the Russian ban on wine (Figure 1). Included in the baseline are higher disbursements of concessional loans over the next four years. An alternative scenario (A2 in Table 1b) that examines the effects of less concessional terms on loans indicates no breach of the thresholds either.

9. **A more worrisome possibility is a fall in transfers.** An alternative scenario (B4 in Table 1b) shows the effects of a shock of one standard deviation below the historical average of net non-debt generating inflows in 2008-09 (which are primarily remittances in Moldova). The thresholds for debt to GDP and debt to exports are breached and do not fall back to more sustainable levels until 2012 and 2014, respectively. This illustrates Moldova's dependence on remittances. The baseline scenario assumes that remittances plateau at the current percent of GDP. However, should the labor environment abroad become less welcoming, or a slowdown in Russia or the EU (the two main destinations) occur, migrants may find it more difficult to maintain their level of support to dependants in Moldova. On the other hand, as remittance flows tend to be much less volatile than other types of inflows, such as portfolio investment, the chances for them to dry up appear slim.[5]

Table 2: Policy-Based External Public Debt Burden Indicators

	Threshold 1/	2007	2008-27 2/
NPV of external debt in percent of:			
Exports	100	51.7	23.6
GDP	30	22.7	9.5
Revenues	200	68.9	26.2
External Debt Service in percent of:			
Exports	15	3.7	1.5
Revenues	25	7.5	2.7

1/ Policy-dependent thresholds as used in the joint IMF-World Bank LIC DSA framework for a poor performer. Moldova's ratings have been improving, but classifications are based on three-year moving averages.
2/ Simple average.

D. Public Debt Sustainability

10. **Public debt and debt-service to revenue ratios remain well below the appropriate thresholds in both the baseline and historical scenarios.** Public domestic debt has remained steady at about 25 percent of total public debt, and is held primarily by the National Bank of Moldova. As there were primary surpluses in recent years, the historical scenario shows how quickly all public debt could be paid off: the debt and debt-service ratios

[5] While the annual growth rate of remittances has been over 30 percent since 2001, experiences from other countries do not imply equal risks of large negative growth. In addition, remittance growth in the first quarter of 2007 is estimated at 29 percent, and the robust growth in the euro area is an upside risk.

all become negative before 2020. In practice, this is unlikely to occur, especially considering Moldova's significant and widely-recognized needs to upgrade infrastructure and make other capital investments to lay the foundation for future growth.

11. **A recession in 2008-09 would derail the baseline scenario.** As shown in Figure 2 (and B1 in Table 2b), a contraction of 1.6 percent in 2008 and 2009 (stemming from a growth shock of one standard deviation below the historical average) would imply debt to GDP above 30 percent by 2011, and continuing to rise afterwards. This scenario assumes that government expenditures continue on their previous trajectory despite the recession. The recession implies that expenditure as a percent of GDP would rise, bringing the deficit to a permanently higher level. Moldova has a relatively high level of government expenditure as a proportion of GDP to begin with, and it has been increasing rapidly in recent years. Arresting this expansion is crucial to maintaining resilience to potential shocks. The growth forecast is good, and Moldova was fortunate that the twin shocks from the wine ban and higher energy prices did not adversely affect growth more. While somewhat extreme, the alternative growth scenario shows how a temporary shock would affect debt sustainability.

E. Conclusions

12. **Moldova's debt outlook is favorable, with a low risk of debt distress.** In the face of some severe shocks in 2006, the Moldovan authorities maintained policies that did not allow the debt situation to deteriorate. This led to an agreement with the Paris Club and commitments by the Consultative Group that have bolstered debt sustainability. Given Moldova's development needs, there appears to be room for modest additional borrowing for infrastructure and other high priority projects, as long as such borrowing remains prudent. Absorptive capacity will likely be a more binding constraint than debt sustainability, and since Moldova is forecast to graduate from IDA-only eligibility in a few years, opportunities for concessional borrowing should be seized in the near term.

13. **Although the risk is low, there are some fragilities.** The external outlook remains highly dependent on remittances, which have expanded rapidly in recent years. An interruption in the ability of migrants to support dependents in Moldova would significantly affect the sustainability of debt. Improving the business climate so that internal growth can attract migrants back into the country would improve resilience. In addition, a temporary recession would cause the public fiscal situation to deteriorate rapidly. A smaller government as a proportion of GDP would be less likely to endanger debt sustainability.

Table 1a. Moldova: External Debt Sustainability Framework, Baseline Scenario, 2007-27 1/
(In percent of GDP, unless otherwise indicated)

	Actual		Historical Average 7/	Standard Deviation 7/	Projections									
	2005	2006			2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
External debt (nominal) 1/	**55.6**	**59.1**	**85.3**	**26.3**	**53.2**	**46.1**	**40.8**	**36.3**	**31.8**	**27.6**	**39.3**	**16.1**	**12.2**	15.4
o/w public and publicly guaranteed (PPG) 2/	27.5	26.0	53.5	26.1	23.3	19.0	16.6	14.7	12.9	11.0	16.3	4.3	0.3	3.3
Change in external debt	-8.2	3.5	-10.8	10.1	-5.9	-7.1	-5.3	-4.5	-4.5	-4.2	-5.2	-0.9	-0.1	-1.0
Identified net debt-creating flows	-5.6	-0.7	-12.3	7.8	-2.4	-3.1	-3.8	-2.9	-3.8	-7.8	-4.0	-7.0	-6.9	-7.0
Non-interest current account deficit	**7.7**	**9.6**	**3.6**	**4.4**	**6.9**	**6.5**	**5.7**	**3.8**	**3.1**	**-1.1**	**4.1**	**-0.8**	**-0.9**	-0.8
Deficit in balance of goods and services	40.5	47.6	30.2	10.2	48.1	47.4	46.3	44.8	42.5	40.0	44.8	40.0	40.0	40.0
Exports	51.1	45.8	50.7	2.4	43.9	42.1	41.2	40.7	40.3	39.8	41.3	39.8	39.8	39.8
Imports	91.7	93.3	81.0	9.2	92.0	89.5	87.4	85.6	82.8	79.8	86.2	79.8	79.8	79.8
Net current transfers (negative = inflow)	-18.2	-23.6	-15.3	4.8	-26.3	-26.3	-25.8	-25.9	-23.9	-25.1	-25.5	-25.1	-25.1	-25.1
Other current account flows (negative = net inflow)	-14.6	-14.3	-11.4	4.8	-14.9	-14.6	-14.7	-15.1	-15.5	-16.0	-15.2	-15.7	-15.7	-15.7
Net FDI (negative = inflow)	**-7.5**	**-6.6**	**-5.3**	**2.4**	**-8.8**	**-9.0**	**-9.0**	**-6.3**	**-6.6**	**-6.4**	**-7.7**	**-6.4**	**-6.4**	-6.4
Endogenous debt dynamics 3/	**-5.8**	**-3.7**	**-10.3**	**5.6**	**-0.5**	**-0.5**	**-0.5**	**-0.3**	**-0.3**	**-0.3**	**-0.4**	**0.1**	**0.4**	0.2
Contribution from nominal interest rate	2.5	2.4	2.9	0.7	2.0	1.8	1.6	1.5	1.3	1.1	1.5	0.8	0.9	0.8
Contribution from real GDP growth	-4.2	-2.0	-5.1	1.9	-2.5	-2.3	-2.0	-1.8	-1.6	-1.4	-2.0	-0.6	-0.5	-0.6
Contribution from price and exchange rate changes	-4.2	-4.1	-8.1	4.9	…	…	…	…	…	…		…	…	…
Residual (3-4) 4/	**-2.6**	**4.2**	1.5	5.2	**-3.5**	**-4.0**	**-1.5**	**-1.7**	**-0.7**	**3.6**	**-1.3**	**6.1**	**6.8**	6.0
o/w exceptional financing	-1.8	-3.4	-2.6	1.3	0.0	0.0	0.0	5.2	0.0	0.0	0.9	0.0	0.0	0.0
NPV of external debt 5/	52.9	59.0	56.8	3.4	52.6	47.8	43.5	39.3	35.4	31.6	41.7	20.5	15.1	19.3
In percent of exports	101.0	128.8	114.2	14.0	119.8	113.5	105.8	96.6	87.9	79.4	100.5	51.4	37.9	48.5
NPV of PPG external debt	**26.4**	**25.8**	27.4	2.3	**22.7**	**20.8**	**19.3**	**17.7**	**16.5**	**15.0**	**18.7**	**8.6**	**3.2**	7.2
In percent of exports	50.3	56.4	54.9	4.0	51.7	49.4	46.9	43.5	41.0	37.6	45.0	21.7	8.0	18.1
Debt service-to-exports ratio (in percent)	20.6	17.7	22.3	3.4	15.9	15.6	15.8	16.0	15.2	13.8	15.4	8.2	8.9	8.8
PPG debt service-to-exports ratio (in percent)	7.0	5.6	11.8	4.6	3.7	3.2	2.8	3.0	2.5	2.7	3.0	1.2	0.6	1.2
Total gross financing need (billions of U.S. dollars)	320.0	372.5	203.4	110.7	200.0	186.5	167.2	231.2	171.3	-145.0	135.2	-460.9	-935.4	-592.1
Non-interest current account deficit that stabilizes debt ratio	15.9	6.2	13.8	8.2	12.8	13.6	11.0	8.3	7.6	3.1	9.4	0.1	-0.8	0.2
Key macroeconomic assumptions														
Real GDP growth (in percent)	7.5	4.0	3.3	4.9	5.0	5.0	5.0	5.0	5.0	5.0	5.0	4.0	4.0	4.0
GDP deflator in U.S. dollar terms (change in percent)	7.0	8.0	10.2	6.3	13.0	10.2	7.8	5.9	5.9	6.1	8.1	4.0	3.0	4.6
Effective interest rate (percent) 6/	4.6	4.8	3.9	0.7	4.0	3.8	3.9	4.2	4.0	3.9	4.0	4.9	7.5	5.7
Growth of exports of G&S (U.S. dollar terms, in percent)	14.8	0.5	15.9	8.6	13.9	11.0	10.6	10.0	10.0	10.0	10.9	8.2	7.1	8.7
Growth of imports of G&S (U.S. dollar terms, in percent)	29.7	14.4	21.8	8.5	16.9	12.6	10.5	8.8	7.6	7.3	10.6	8.2	7.1	8.7
Grant element of new public sector borrowing (in percent)	...	...	...	...	37.1	36.9	34.4	39.3	31.4	27.8	34.5	14.6	14.6	14.6
Memorandum item:														
Nominal GDP (millions of U.S. dollars)	2988.3	3356.2			3982.8	4606.7	5213.3	5797.5	6444.3	7178.4	5225.6	11823.2	25166.0	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Does not include debt of public enterprises.
3/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
4/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
5/ Assumes that NPV of private sector debt is equivalent to its face value.
6/ Current-year interest payments devided by previous period debt stock.
7/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 1b. Moldova: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007-27 1/
(In percent)

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of debt-to-GDP ratio								
Baseline	23	21	19	18	17	15	9	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-27 2/	23	21	20	18	17	19	26	28
A2. New public sector loans on less favorable terms in 2007-27 3/	23	21	20	19	18	16	10	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	23	22	22	20	19	17	10	4
B2. Export value growth at historical average minus one standard deviation in 2007-08 4/	23	22	22	21	19	18	10	3
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2007-08	23	22	21	19	18	16	9	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 5/	23	36	49	45	42	39	20	4
B5. Combination of B1-B4 using one-half standard deviation shocks	23	33	43	40	37	34	18	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 6/	23	29	27	24	23	21	12	4
NPV of debt-to-exports ratio								
Baseline	52	49	47	43	41	38	22	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-27 2/	52	50	49	44	43	48	65	71
A2. New public sector loans on less favorable terms in 2007-27 3/	52	50	49	46	44	41	25	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	52	49	47	43	41	38	22	8
B2. Export value growth at historical average minus one standard deviation in 2007-08 4/	52	54	58	54	51	47	26	9
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2007-08	52	49	47	43	41	38	22	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 5/	52	86	118	111	105	98	51	10
B5. Combination of B1-B4 using one-half standard deviation shocks	52	72	91	86	81	76	40	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 6/	52	49	47	43	41	38	22	8
Debt service ratio								
Baseline	4	3	3	3	3	3	1	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-27 2/	4	3	3	3	3	3	3	6
A2. New public sector loans on less favorable terms in 2007-27 3/	4	3	3	3	3	3	1	1
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	4	3	3	3	3	3	2	1
B2. Export value growth at historical average minus one standard deviation in 2007-08 4/	4	3	3	4	3	4	2	1
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2007-08	4	3	3	3	3	3	2	1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 5/	4	3	4	5	5	5	6	2
B5. Combination of B1-B4 using one-half standard deviation shocks	4	3	3	4	4	4	4	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 6/	4	3	3	3	3	3	2	1
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 7/	19	19	19	19	19	19	19	19

Source: Staff projections and simulations.

1/ Does not include debt of public enterprises.
2/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
4/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
5/ Includes official and private transfers and FDI.
6/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
7/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Moldova: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004-27
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2004	2005	2006	Historical Average 5/	Standard Deviation 5/	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Public sector debt 1/	47.2	38.0	33.0	44.0	32.8	28.9	25.9	22.9	19.9	17.6	15.4	21.8	7.9	3.3	6.8
o/w foreign-currency denominated	35.6	27.9	24.4	34.8	29.7	21.1	18.9	16.7	14.9	13.4	11.8	16.1	6.1	2.1	5.1
Change in public sector debt	-13.1	-9.2	-5.0	3.3	27.1	-4.0	-3.0	-3.0	-3.0	-2.2	-2.2	-2.9	-0.8	-0.3	-0.8
Identified debt-creating flows	-11.6	-7.8	-5.8	-11.4	5.3	-5.0	-3.1	-2.4	-1.9	-1.8	-1.7	-2.7	0.3	1.5	0.6
Primary deficit	-2.6	-2.8	-1.3	-0.7	3.5	-0.8	-0.6	-0.5	-0.5	-0.5	-0.5	-0.6	0.8	1.7	1.1
Revenue and grants	35.4	38.6	40.5	34.5	4.0	42.0	39.7	39.5	39.5	39.5	39.5	39.9	38.2	37.3	37.9
of which : grants	0.4	1.2	0.7	0.7	0.3	2.1	2.6	2.8	2.8	2.8	2.8	2.6	1.5	0.6	1.2
Primary (noninterest) expenditure	32.7	35.8	39.2	33.8	6.3	41.2	39.1	39.0	39.0	39.0	39.0	39.4	39.0	39.0	39.0
Automatic debt dynamics	-8.9	-4.9	-4.1	-8.8	4.4	-3.7	-2.5	-1.8	-1.4	-1.3	-1.1	-2.0	-0.5	-0.2	-0.4
Contribution from interest rate/growth differential	-4.2	-3.9	-2.2	-4.1	4.6	-1.2	-1.2	-1.0	-0.9	-0.8	-0.7	-1.0	-0.3	-0.1	-0.2
of which : contribution from average real interest rate	0.0	-0.6	-0.7	0.0	0.9	0.4	0.2	0.2	0.2	0.1	0.1	0.2	0.1	0.0	0.1
of which : contribution from real GDP growth	-4.2	-3.3	-1.5	-2.5	2.4	-1.6	-1.4	-1.2	-1.1	-0.9	-0.8	-1.2	-0.3	-0.1	-0.3
Contribution from real exchange rate depreciation	-4.7	-1.0	-1.9	-3.4	3.0	-2.5	-1.3	-0.8	-0.4	-0.5	-0.4	-1.0	...	...	-0.3
Other identified debt-creating flows	-0.1	-0.1	-0.4	-0.9	1.0	-0.4	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	0.0	0.0	0.0
Privatization receipts (negative)	-0.1	-0.1	-0.4	-0.9	1.0	-0.4	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	0.0	0.0	0.0
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Residual, including asset changes	-1.5	-1.4	0.7	1.7	3.3	0.9	0.1	-0.7	-1.1	-0.4	-0.5	-0.3	-1.0	-1.8	-1.4
NPV of public sector debt	41.5	34.6	27.4	40.9	31.7	22.8	18.8	17.5	15.2	13.8	12.5	16.8	8.1	3.5	6.7
o/w foreign-currency denominated	29.9	24.6	18.8	31.8	28.7	14.9	11.8	11.2	10.3	9.6	8.8	11.1	6.3	2.3	5.0
o/w external	29.9	24.6	18.8	45.4	22.8	14.9	11.8	11.2	10.3	9.6	8.8	11.1	6.3	2.3	5.0
NPV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Gross financing need 2/	-0.8	0.8	5.8	3.5	4.0	6.2	6.5	6.5	5.7	3.7	3.3	5.3	3.0	3.1	3.2
NPV of public sector debt-to-revenue ratio (in percent) 3/	117.3	89.9	67.6	127.4	107.3	54.2	47.4	44.2	38.6	35.0	31.6	41.8	21.1	9.5	17.7
o/w external	84.5	63.8	46.4	142.1	83.9	35.5	29.8	28.5	26.1	24.2	22.3	27.7	16.4	6.2	13.2
Debt service-to-revenue ratio (in percent) 3/ 4/	5.4	8.6	8.8	11.8	6.0	7.5	6.7	5.9	4.3	3.8	3.5	5.3	1.4	1.0	1.6
Primary deficit that stabilizes the debt-to-GDP ratio	10.5	6.4	3.7	6.9	3.4	3.2	2.4	2.5	2.5	1.7	1.7	2.3	1.6	1.9	1.9
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.4	7.5	4.0	3.3	4.9	5.0	5.0	5.0	5.0	5.0	5.0	5.0	4.0	4.0	4.0
Average nominal interest rate on forex debt (in percent)	2.5	3.0	2.7	3.2	0.6	1.6	1.5	2.4	2.3	2.3	2.2	2.1	2.3	2.6	2.4
Average real interest rate on domestic currency debt (in percent)	1.1	-5.3	-7.2	-3.6	5.2	4.7	3.7	2.4	1.9	1.9	1.9	2.8	1.9	1.9	1.9
Real exchange rate depreciation (in percent, + indicates depreciation)	-10.1	-3.0	-7.0	1.2	24.5	-10.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.0	9.3	12.5	15.6	10.1	7.4	7.4	6.8	5.7	5.7	5.7	6.5	5.7	5.7	5.7
Growth of real primary spending (deflated by GDP deflator, in percent)	12.6	17.5	14.0	4.1	16.8	10.3	-0.3	4.7	5.0	5.0	5.0	5.0	4.0	4.0	4.0
Grant element of new external borrowing (in percent)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...

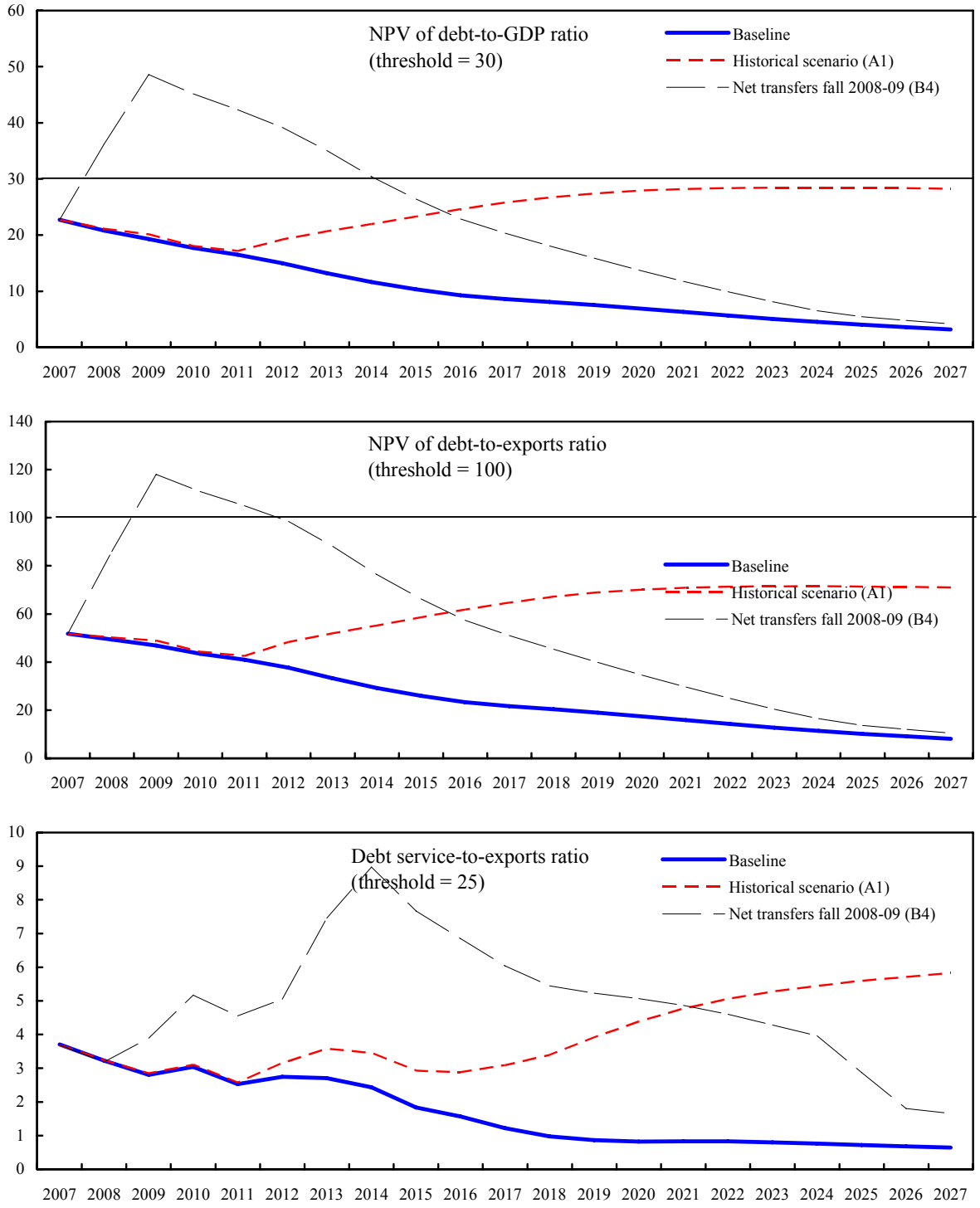
Sources: Moldovan authorities; and Fund staff estimates and projections.
1/ General government gross debt, excluding debt of public enterprises.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues including grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b. Moldova: Sensitivity Analysis for Key Indicators of Public Debt 2007-27

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of Debt-to-GDP Ratio								
Baseline	23	19	17	15	14	12	8	4
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	23	19	18	16	14	13	3	-17
A2. Primary balance is unchanged from 2007	23	19	17	14	13	11	2	-17
A3. Permanently lower GDP growth 1/	23	19	19	18	18	18	27	69
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-09	23	22	27	29	31	34	46	64
B2. Primary balance is at historical average minus one standard deviations in 2008-09	23	22	23	21	19	17	12	6
B3. Combination of B1-B2 using one half standard deviation shocks	23	21	22	19	17	15	9	3
B4. One-time 30 percent real depreciation in 2008	23	24	22	20	18	17	12	6
B5. 10 percent of GDP increase in other debt-creating flows in 2008	23	28	26	23	22	20	14	7
NPV of Debt-to-Revenue Ratio 2/								
Baseline	54	47	44	39	35	32	21	9
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	54	48	45	40	36	33	9	-45
A2. Primary balance is unchanged from 2007	54	47	43	37	32	28	4	-45
A3. Permanently lower GDP growth 1/	54	49	48	45	45	46	71	186
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-09	54	56	68	72	78	84	119	171
B2. Primary balance is at historical average minus one standard deviations in 2008-09	54	55	59	52	48	44	31	15
B3. Combination of B1-B2 using one half standard deviation shocks	54	53	55	48	43	39	25	8
B4. One-time 30 percent real depreciation in 2008	54	60	57	51	47	43	32	15
B5. 10 percent of GDP increase in other debt-creating flows in 2008	54	71	66	59	55	50	36	18
Debt Service-to-Revenue Ratio 2/								
Baseline	8	7	6	4	4	3	1	1
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	7	6	4	4	4	0	-5
A2. Primary balance is unchanged from 2007	8	7	6	4	3	3	-1	-5
A3. Permanently lower GDP growth 1/	8	7	6	5	5	6	7	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-09	8	7	9	10	11	11	12	20
B2. Primary balance is at historical average minus one standard deviations in 2008-09	8	7	9	8	5	4	2	2
B3. Combination of B1-B2 using one half standard deviation shocks	8	7	8	6	5	4	1	1
B4. One-time 30 percent real depreciation in 2008	8	7	6	5	4	4	2	1
B5. 10 percent of GDP increase in other debt-creating flows in 2008	8	7	14	8	6	5	2	2

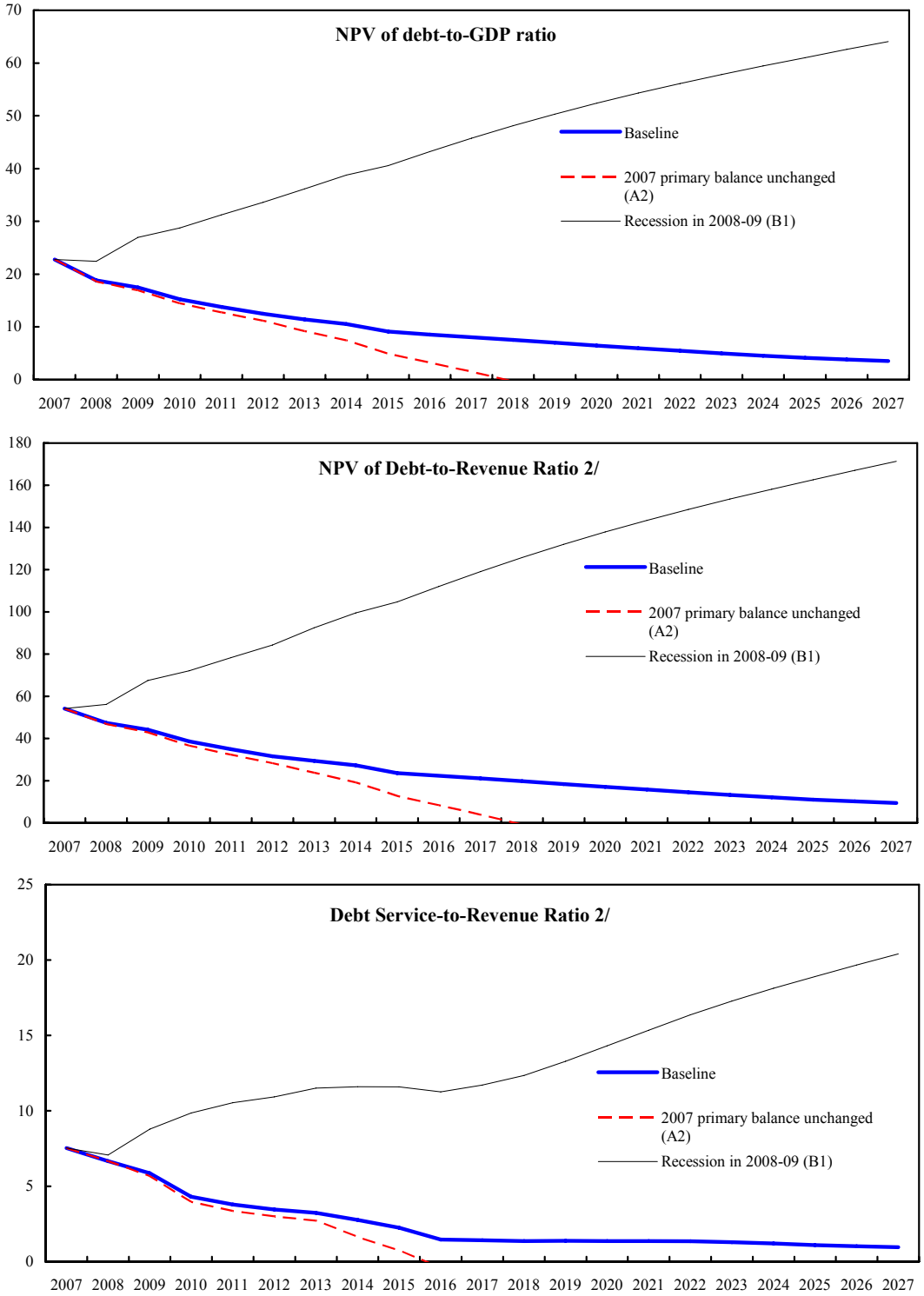
Sources: Moldovan authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Figure 1. Moldova: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2007-27
(In percent)



Source: Staff projections and simulations.

Figure 2. Moldova: Indicators of Public Debt Under Alternative Scenarios, 2007-27 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2017.
2/ Revenue including grants.